Lecture 4

The linear Cauchy problem

The field equation from Lecture 1 does not by itself construct a field from initial data. We now solve the Klein–Gordon equation on all of $\mathbb{R}^d$. Fourier transformation replaces the spatial derivatives by oscillator frequencies.

Objectives. *We construct the solution from finite-energy data and prove uniqueness, continuous dependence and conservation of total energy, including zero mass.*

By the end of this lecture you will be able to:

- Define the energy space by Fourier transformation.
- Construct and identify the unique energy-class Klein–Gordon solution.
- Control its norm on bounded time intervals, including at zero frequency.

4.1 Data and the solution class

The unitary Fourier transform from *Analysis* I, Lecture 7, has kernel $(2\pi)^{-d/2}e^{-ik\cdot x}$. It maps $L^2(\mathbb{R}^d)$ onto itself, preserves norms, and takes a Schwartz derivative to $ik_j\widehat{f}$. Write $\langle k\rangle = (1+|k|^2)^{1/2}$.

Definition 4.1 (Fourier energy space)**.** The *Sobolev space* $H^1(\mathbb{R}^d)$ and its norm are

$$H^1(\mathbb{R}^d) = \{f \in L^2(\mathbb{R}^d) : \langle k\rangle \widehat{f} \in L^2(\mathbb{R}^d)\}, \qquad \|f\|_{H^1} = \|\langle k\rangle \widehat{f}\|_2.$$

The *energy space* is $\mathcal{E} = H^1(\mathbb{R}^d) \times L^2(\mathbb{R}^d)$, with norm $\|(f,g)\|_{\mathcal{E}} = \|f\|_{H^1} + \|g\|_2$.

For $f \in H^1$, define $\partial_j f$ as the inverse transform of $ik_j\widehat{f}$. Plancherel gives

$$\|f\|_{H^1}^2 = \|f\|_2^2 + \sum_{j=1}^{d} \|\partial_j f\|_2^2.$$

These derivatives agree with distributional derivatives: Fourier inversion on Schwartz tests gives $\int (\partial_j f)\zeta = -\int f\partial_j\zeta$. All pairings here are bilinear. The integrals exist by Cauchy–Schwarz. The weighted L^2 definition also proves completeness of H^1. Indeed, if $\langle k\rangle\widehat{f_n} \to h$ in L^2, then $\widehat{f_n} \to h/\langle k\rangle$ in L^2, whose inverse transform belongs to H^1 and is the required limit.

We consider real data. Fourier transforms may be complex, with $\widehat{f}(-k) = \overline{\widehat{f}(k)}$. Fix a mass $m \geq 0$ and an open time interval I containing zero.

Definition 4.2 (Energy-class solution and well-posedness)**.** An *energy-class solution* with data $(f,g)\in\mathcal{E}$ is a map $v\in C(I;H^1)\cap C^1(I;L^2)$ satisfying $v(0)=f$ and $v_t(0)=g$, and the equation

$$\int_I\int_{\mathbb{R}^d} v(t,x)\bigl(\psi_{tt}-\Delta\psi+m^2\psi\bigr)(t,x)\,\mathrm{d}^dx\,\mathrm{d}t=0\quad\text{for every }\psi\in C_c^\infty(I\times\mathbb{R}^d). \tag{4.1}$$

The Cauchy problem on $\mathbb{R}$ is *well-posed* in $\mathcal{E}$ if each datum has exactly one energy-class solution with $I=\mathbb{R}$ and the map from data to $(v,v_t)\in C([-T,T];\mathcal{E})$ is continuous for every $T>0$.

The time derivative in this definition is a derivative in the L^2 norm. Equation (4.1) defines the distributional meaning of $v_{tt}-\Delta v+m^2v=0$. It requires neither pointwise second derivatives nor products of distributions.

4.2 Construction by Fourier multipliers

The oscillator with initial position a and velocity b solves $q''+\nu^2q=0$. Its solution is $q(t)=\cos(t\nu)a+S_t(\nu)b$, where the *sine multiplier* is the continuous function

$$S_t(\nu)=\begin{cases}\sin(t\nu)/\nu, & \nu>0,\\ t, & \nu=0.\end{cases}\qquad |S_t(\nu)|\le|t|,\qquad |\nu S_t(\nu)|\le 1. \tag{4.2}$$

The integral formula $S_t(\nu)=\int_0^t\cos(s\nu)\,\mathrm{d}s$ proves the first bound and the value at zero. The second bound follows from $|\sin|\le 1$. The transformed Klein–Gordon frequency is $\nu(k)=(|k|^2+m^2)^{1/2}$.

Theorem 4.3 (Klein–Gordon Cauchy problem)**.** *For every $(f,g)\in\mathcal{E}$ and $m\ge 0$, there is exactly one energy-class solution on $\mathbb{R}$. It is given by*

$$\widehat{v}(t,k)=\cos(t\nu(k))\widehat{f}(k)+S_t(\nu(k))\widehat{g}(k). \tag{4.3}$$

The problem is well-posed. The total energy is constant:

$$E_{\mathrm{en}}(t)=\frac{1}{2}\bigl(\|v_t(t)\|_2^2+\|\nabla v(t)\|_2^2+m^2\|v(t)\|_2^2\bigr)=E_{\mathrm{en}}(0). \tag{4.4}$$

Here $\|\nabla v\|_2^2=\sum_j\|\partial_j v\|_2^2$.

Proof. We first construct the solution and justify its time regularity. For $|t|\le T$, the multiplier bounds give

$$\begin{aligned}\|v(t)\|_2&\le\|f\|_2+T\|g\|_2,\\ \|v(t)\|_{H^1}&\le\|f\|_{H^1}+(T+1)\|g\|_2,\\ \|p(t)\|_2&\le\max(1,m)\|f\|_{H^1}+\|g\|_2,\end{aligned} \tag{4.5}$$

where $\widehat{p}=-\nu\sin(t\nu)\widehat{f}+\cos(t\nu)\widehat{g}$. For the middle bound, use $\langle k\rangle|S_t|\le(1+|k|)|S_t|\le T+1$. For the last bound, use $\nu\le\max(1,m)\langle k\rangle$.

Every multiplier is pointwise continuous in time. The same bounds supply square-integrable majorants for $\langle k\rangle\widehat{v}$ and $\widehat{p}$ on each bounded time interval. Dominated convergence, imported from *Analysis* I, Lecture 3, gives $v\in C(\mathbb{R};H^1)$ and $p\in C(\mathbb{R};L^2)$. At each k, the

difference quotient of $\widehat{v}$ tends to $\widehat{p}$. Its absolute value, for $|h| \le 1$, is bounded by $\nu|\widehat{f}| + |\widehat{g}|$, by the scalar mean-value estimate. Dominated convergence in L^2 therefore gives $v_t = p$. The values at zero are f and g. Evenness and reality of the multipliers preserve the Fourier reality condition.

To verify the equation, choose an even real $\chi \in C_c^\infty(\mathbb{R}^d)$ equal to one on $|k| \le 1$, supported in $|k| \le 2$, with $0 \le \chi \le 1$. Write $\chi_R(k) = \chi(k/R)$ for $R \ge 1$. Multiply both data transforms by χ_R and construct v_R by the same formula. Compact frequency support and Cauchy–Schwarz make every polynomial in k times either truncated transform integrable. Differentiation under the inverse integral is therefore valid to every order, also in time on compact intervals. The oscillator equation proves that v_R is a smooth classical solution. It satisfies (4.1) by integration by parts against the compact test. The estimates (4.5) and dominated convergence give

$$v_R \longrightarrow v \text{ in } C([-T,T];H^1), \qquad (v_R)_t \longrightarrow v_t \text{ in } C([-T,T];L^2). \tag{4.6}$$

Cauchy–Schwarz on the test's compact time support passes the weak equation to the limit. This also establishes the smooth approximation used in Lecture 5.

Next we prove uniqueness within the stated solution class. Let u be the difference of two solutions with the same data. Set $w_R = \mathcal{F}^{-1}(\chi_R \widehat{u})$. We justify the equation for this cutoff before using an oscillator argument. The weak equation permits tests $\eta(t)\zeta(x)$ with $\eta \in C_c^\infty(I)$ and ζ Schwartz: multiply ζ by a spatial cutoff tending to one. The test and its first two spatial derivatives converge in L^2, so Cauchy–Schwarz passes to the limit. The Fourier transform takes Schwartz space onto itself. Testing with inverse transforms of χ_R times arbitrary Schwartz functions consequently gives, as an L^2-valued distribution in time,

$$(w_R)_{tt} = -Aw_R, \qquad \widehat{Aw_R}(k) = (|k|^2 + m^2)\widehat{w}_R(k).$$

Indeed, scalar equality against the dense Schwartz tests identifies the L^2 vectors after integration against each η. The operator A is bounded on functions supported in $|k| \le 2R$ in Fourier space. Thus $-Aw_R$ is continuous in L^2.

This equation implies $w_R \in C^2(I;L^2)$. To see this, integrate $-Aw_R$ from zero in the L^2 norm. The integral exists as a limit of Riemann sums because L^2 is complete. Subtract this primitive from $(w_R)_t$. Each scalar pairing of the difference has zero distributional derivative, so is constant. The scalar assertion follows by writing every test of integral zero as a derivative of a compactly supported test. Evaluation at zero identifies the constant. Thus $(w_R)_t$ is norm differentiable, with derivative $-Aw_R$. The cutoff now has classical time derivatives, and differentiation gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(\|(w_R)_t\|_2^2 + \|\nu\widehat{w}_R\|_2^2\big) = 2\,\mathrm{Re}\big(((w_R)_t, -Aw_R) + (Aw_R, (w_R)_t)\big) = 0.$$

The inner product is antilinear in its first slot. The initial value is zero, so $(w_R)_t = 0$ and then $w_R = w_R(0) = 0$. As $R \to \infty$, Plancherel and dominated convergence give $u(t) = 0$. This proves uniqueness without assuming a pointwise Fourier equation for u.

For continuous dependence, apply (4.5) to the difference of two data pairs. Taking the supremum over $|t| \le T$ gives

$$\sup_{|t|\le T} \|(v-\widetilde{v}, v_t-\widetilde{v}_t)(t)\|_{\mathcal{E}} \le C_{T,m}\|(f-\widetilde{f}, g-\widetilde{g})\|_{\mathcal{E}}, \quad C_{T,m} = T + 2 + \max(1,m).$$

This is continuity in the required norm.

Finally, the formula for the solution and its derivative reads

$$\nu\widehat{v} = \cos(t\nu)\nu\widehat{f} + \sin(t\nu)\widehat{g},$$
$$\widehat{v}_t = -\sin(t\nu)\nu\widehat{f} + \cos(t\nu)\widehat{g}.$$

Expansion of the absolute squares cancels their opposite cross terms and uses $\cos^2(t\nu) + \sin^2(t\nu) = 1$. Hence, for almost every k,

$$|\widehat{v}_t|^2 + \nu^2|\widehat{v}|^2 = |\widehat{g}|^2 + \nu^2|\widehat{f}|^2.$$

Every term is integrable by (4.5). Integration and Plancherel prove (4.4). ∎

For smooth solutions this is the energy of Lecture 3 with $V(v) = \frac{1}{2}m^2v^2$. That lecture proved a local balance for C^2 fields. The multiplier proof above establishes the global identity in the larger energy class, without assuming a boundary flux at spatial infinity.

4.3 A Gaussian and the low-frequency guard

Take one space dimension and Gaussian displacement with zero velocity:

$$\widehat{f}(k) = e^{-ak^2/2}, \qquad g = 0, \qquad a > 0.$$

The Gaussian transform from *Analysis* I, Lecture 4, gives $f(x) = a^{-1/2}e^{-x^2/(2a)}$. Its solution and initial energy are

$$\widehat{v}(t,k) = \cos\bigl(t\sqrt{k^2+m^2}\bigr)e^{-ak^2/2}, \qquad E_{\mathrm{en}}(0) = \frac{\sqrt{\pi}}{4a^{3/2}} + \frac{m^2\sqrt{\pi}}{2\sqrt{a}}.$$

Indeed, the Gaussian integral and its derivative with respect to a give $\int e^{-ak^2}\mathrm{d}k = \sqrt{\pi/a}$ and $\int k^2e^{-ak^2}\mathrm{d}k = \sqrt{\pi}/(2a^{3/2})$. For $m = 0$, the identity $\cos(t|k|) = \cos(tk)$ splits the inverse transform into two translated Gaussians:

$$v(t,x) = \frac{1}{2\sqrt{a}}\left(e^{-(x-t)^2/(2a)} + e^{-(x+t)^2/(2a)}\right).$$

The two profiles travel at unit speed. Their sum has the same total energy at every time even when they overlap.

The massless energy does not bound the full energy-space norm. For an explicit low-frequency family, take $m = 0$, $g_\varepsilon = 0$ and

$$\widehat{f_\varepsilon}(k) = \varepsilon^{-1/2}e^{-k^2/(2\varepsilon^2)}, \qquad 0 < \varepsilon \le 1.$$

Substitution $k = \varepsilon q$ gives

$$\|f_\varepsilon\|_2^2 = \sqrt{\pi}, \qquad E_{\mathrm{en}}[f_\varepsilon, 0] = \frac{\sqrt{\pi}}{4}\varepsilon^2 \longrightarrow 0.$$

No constant can bound $\|f\|_2^2$ by the massless energy for all data. The bound $\|v(t)\|_2 \le \|f\|_2 + |t|\|g\|_2$ must therefore be retained. The value $S_t(0) = t$ also shows why division by the frequency cannot be estimated by a uniform bound on $1/\nu$. Lecture 15 uses this finite-time control when it adds a nonlinear forcing term.

Summary. For $m \ge 0$, data in $H^1(\mathbb{R}^d) \times L^2(\mathbb{R}^d)$ determine exactly one solution in $C(H^1) \cap C^1(L^2)$ of the distributional Klein–Gordon equation. Fourier multipliers give continuous

dependence on bounded time intervals, smooth approximation and conserved energy. At zero mass, energy omits control of the low-frequency L^2 norm. The sine bound supplies that control for finite time. Lecture 5 uses the smooth approximation to prove finite speed of propagation.

Self-check. At $m = 0$ and $k = 0$, which oscillator solution has initial position a and velocity b? How does the multiplier formula reproduce it?

Answer. The equation is $q'' = 0$, so $q(t) = a + tb$. The values $\cos 0 = 1$ and $S_t(0) = t$ give exactly this solution. A single point in frequency space has measure zero, but the continuous extension controls frequencies approaching it.

Self-check. For $m > 0$, bound $\|v(t)\|_2$ by the conserved energy. Why does the resulting bound fail at $m = 0$?

Answer. The mass term gives $\|v(t)\|_2 \le \sqrt{2E_{\mathrm{en}}(0)}/m$. At zero mass this term vanishes. The family f_ε above has fixed nonzero L^2 norm and energy tending to zero.

Self-check. For the Gaussian displacement, compute $\|f\|_{H^1}^2$ and the massless solution's value at $x = 0$.

Answer. The norm is $\sqrt{\pi}/\sqrt{a} + \sqrt{\pi}/(2a^{3/2})$. The value is $v(t, 0) = a^{-1/2} e^{-t^2/(2a)}$.

Self-check. Does the definition of an energy-class solution permit differentiating its energy directly as in Lecture 3? Which step in the uniqueness proof makes time differentiation legitimate?

Answer. The definition supplies no pointwise second derivatives. The frequency cutoff makes the spatial operator bounded in L^2. Its distributional equation then implies C^2 time regularity in L^2, so the cutoff energy can be differentiated. The total energy of the original solution follows instead from the integrable Fourier identity.